VOUGNE FROID ALIS

RECURSION & PROBLEM SOLVING

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WHAT IS RECURSION?

Definition: The process in which a function calls itself directly or indirectly is called recursion and the corresponding function is called a recursive function.

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THE RULES OF RECURSION

1. The Base Case (The Brake Pedal)
The condition where the function stops calling itself.
Example: "If the number is 1, stop."

2. The Recursive Case (The Gas Pedal)
The part where the function calls itself with a smaller input.
Example: "Return n * factorial(n-1)."

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LET’S VISUALIZE!!

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EXAMPLE:

Write a recursive function that given an input n sums all nonnegative integers up to n

1. What is the simplest possible input

```
sum(0) = 0
sum(1) = 1
sum(4) = 1+2+3+4      = 10
sum(5) = 1+2+3+4+5    = 15
```

Lets take a look at the iterative approach

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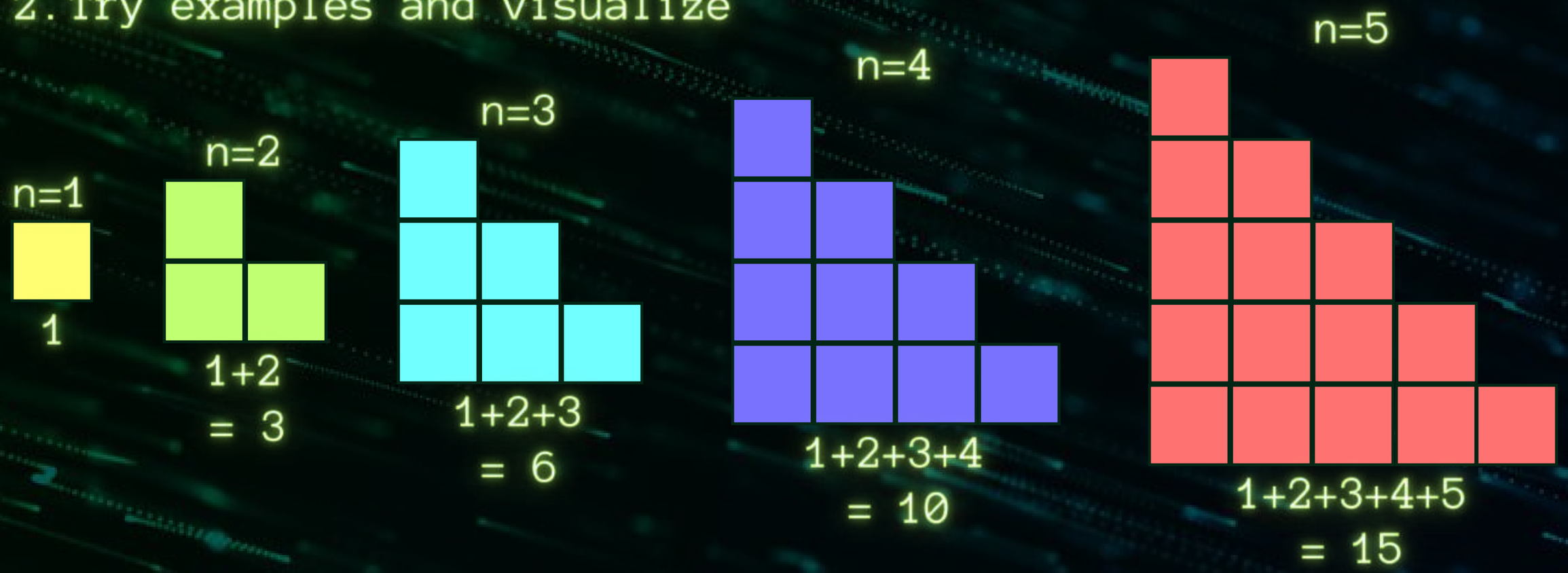
EXAMPLE:

Write a recursive function that given an input n sums all nonnegative integers up to n

2. Try examples and visualize



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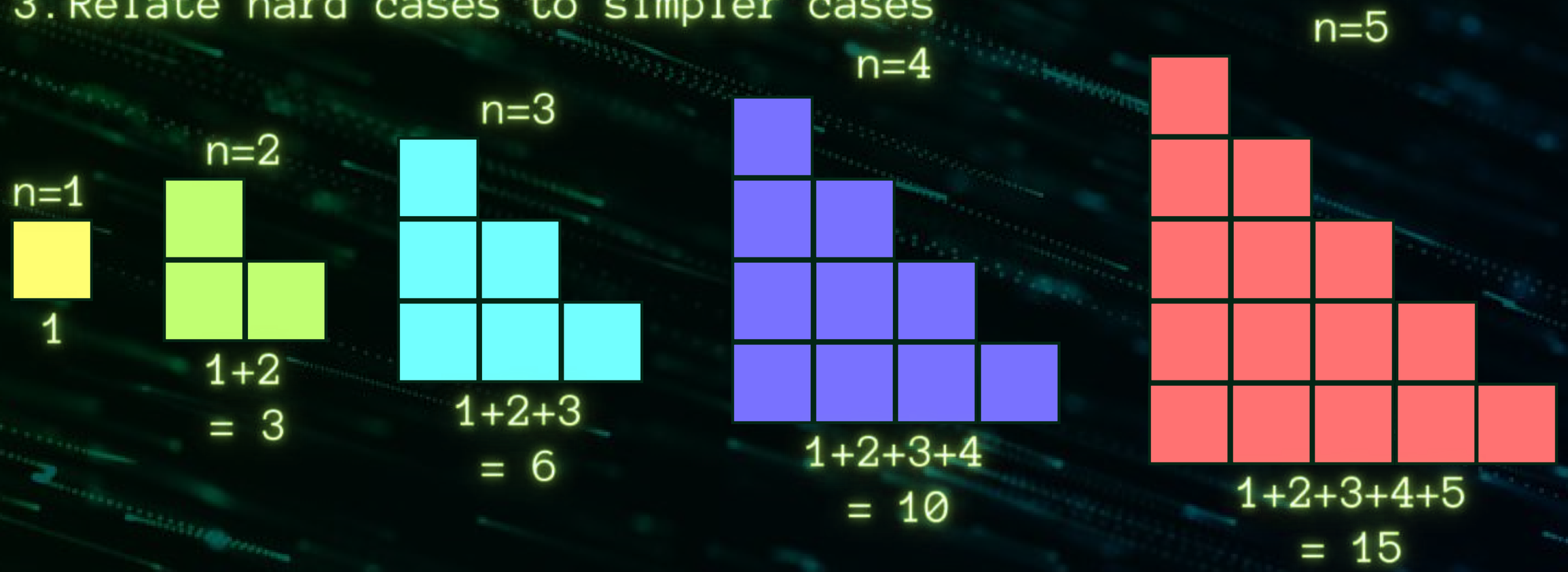
EXAMPLE:

Write a recursive function that given an input n sums all nonnegative integers up to n

3. Relate hard cases to simpler cases



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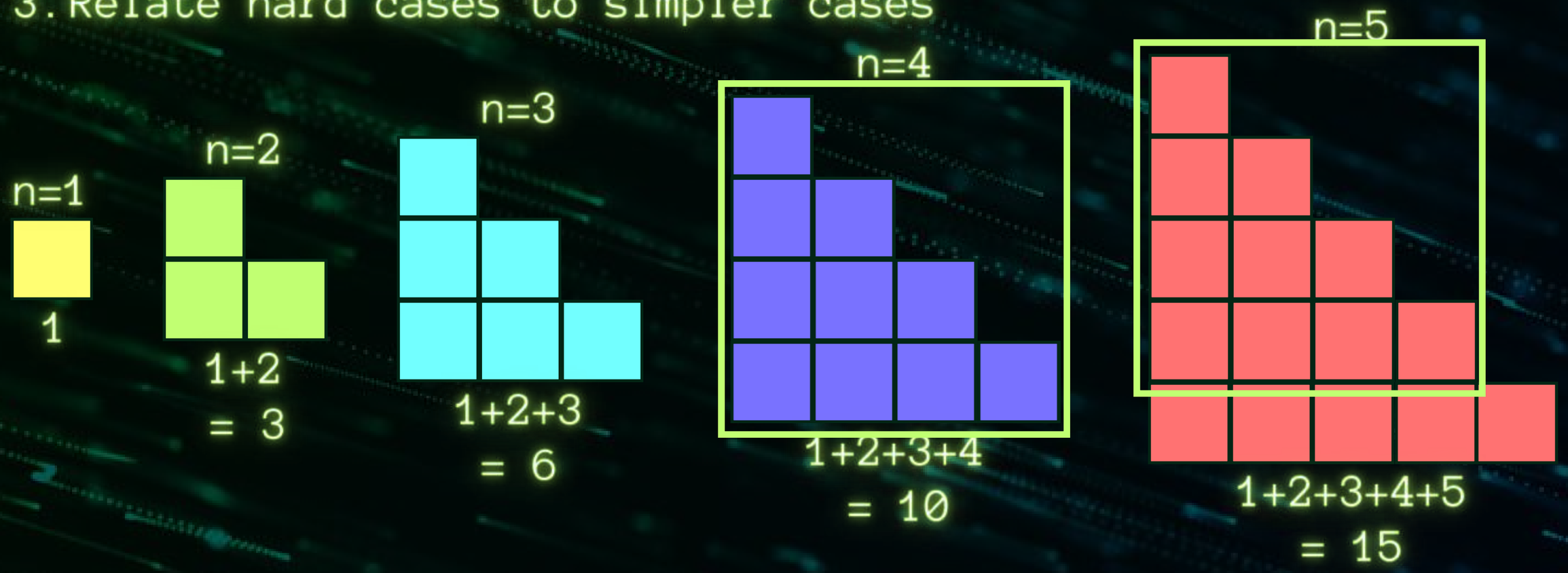

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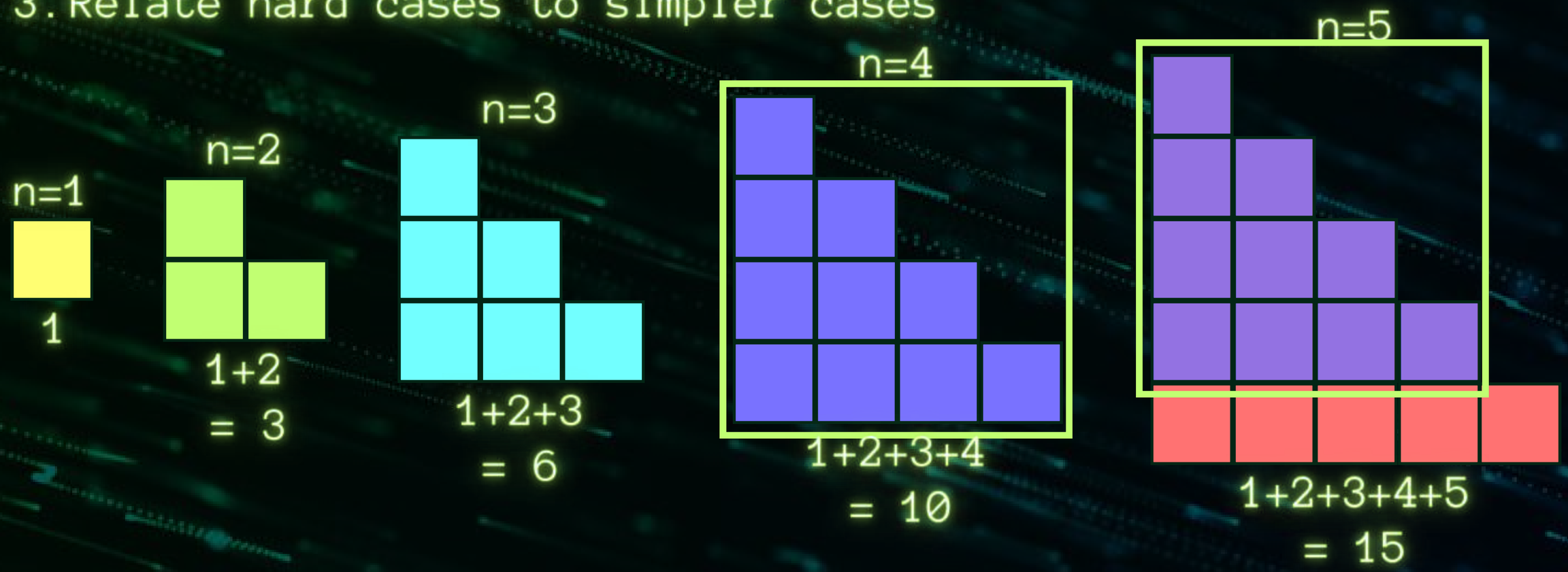

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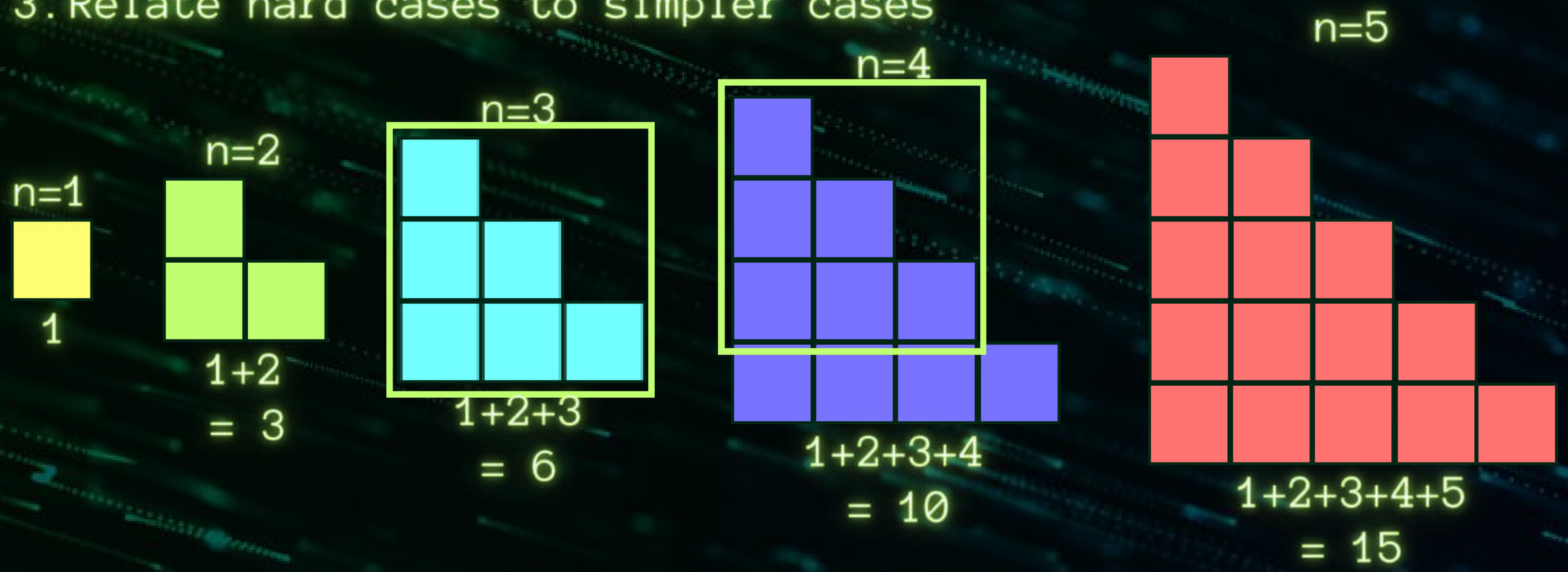

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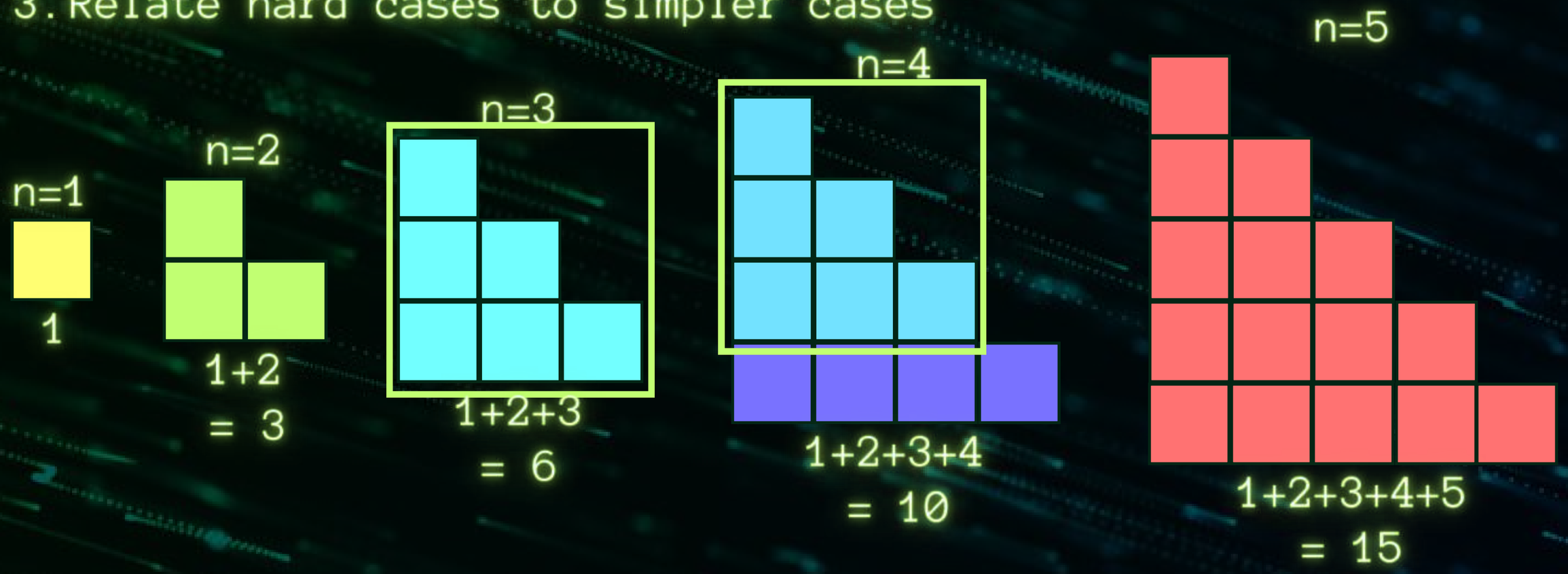

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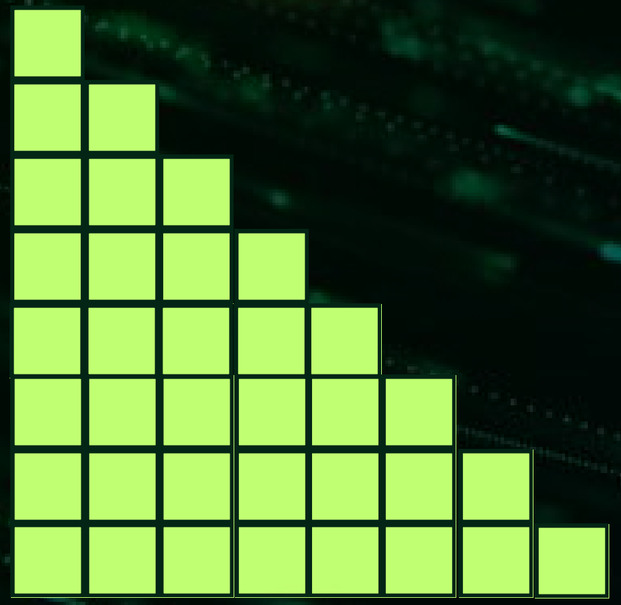
EXAMPLE:

Write a recursive function that given an input n sums all nonnegative integers up to n

4. Generalize the pattern

n=k



=

k

1+2+...+k

sum(k)

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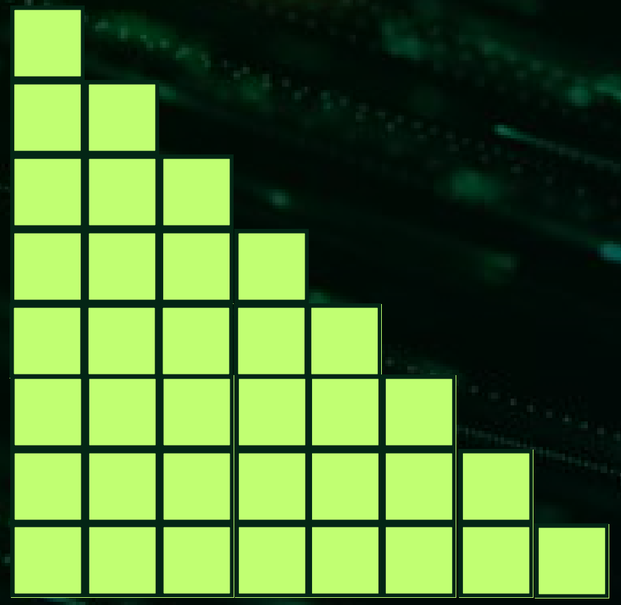
Write a recursive function that given an input n sums all nonnegative integers up to n

4. Generalize the pattern

n=k



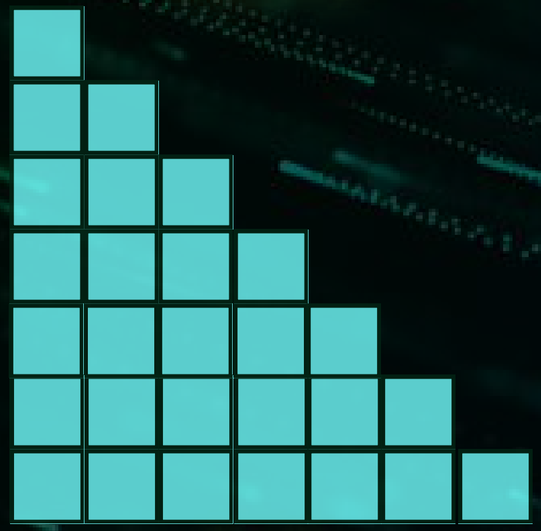
k

1+2+...+k

sum(k)

=

n=k-1



k-1

1+2+...+(k-1)

sum(k-1)

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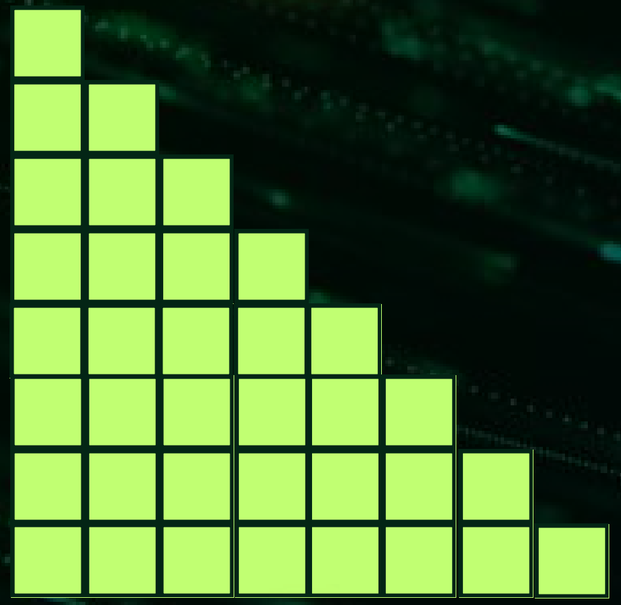
EXAMPLE:

Write a recursive function that given an input n sums all nonnegative integers up to n

4. Generalize the pattern

n=k



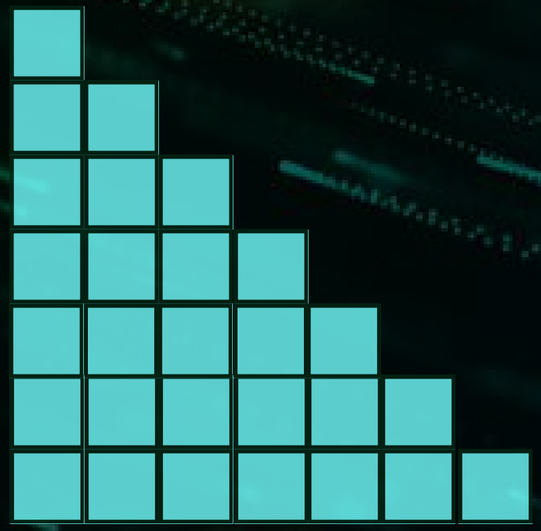
k

1+2+...+k

sum(k)

=

n=k-1



k-1

1+2+...+(k-1)

sum(k-1)

+

k

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EXAMPLE:

Write a recursive function that given an input n sums all nonnegative integers up to n

5. Write the code by combining the recursive pattern with the base case

$$sum(k) = sum(k-1) + k$$

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EXAMPLE:

Write a recursive function that given an input n sums all nonnegative integers up to n

How does the code work?

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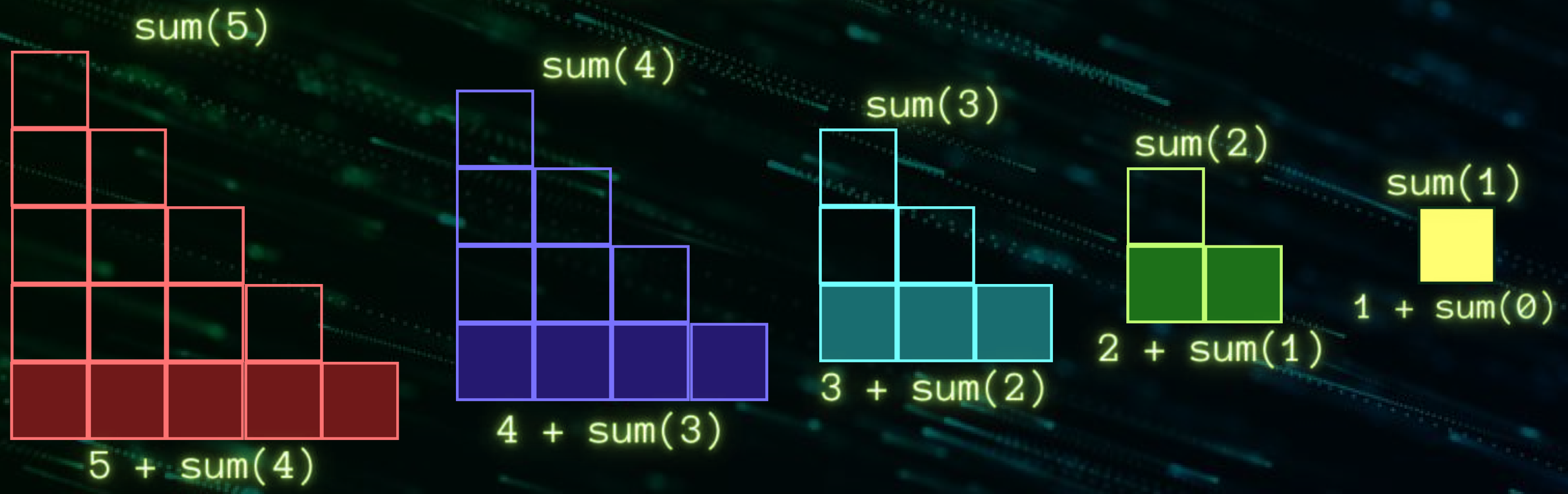

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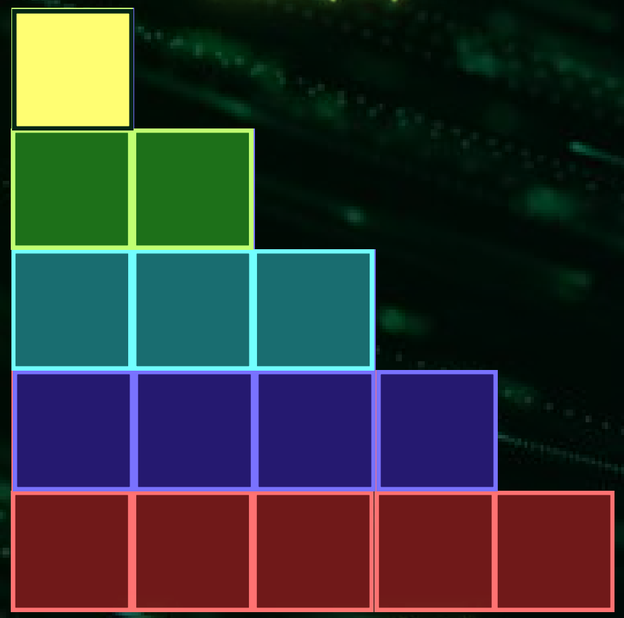
EXAMPLE:

Write a recursive function that given an input n sums all nonnegative integers up to n

How does the code work?

sum(5)

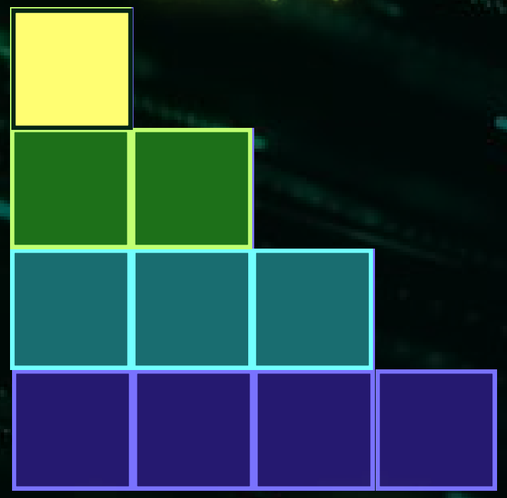


5 + sum(4)

5 + 10

15

sum(4)

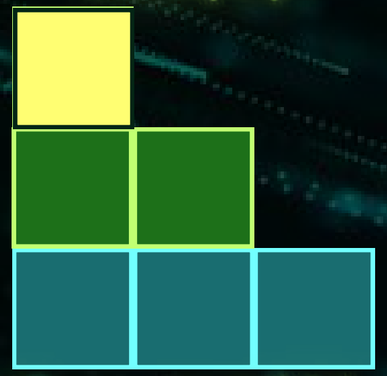


4 + sum(3)

4 + 6

10

sum(3)

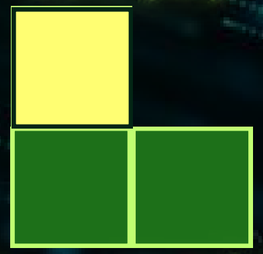


3 + sum(2)

3 + 3

5

sum(2)



2 + sum(1)

2 + 1

3

sum(1)

1 + sum(0)

1 + 0

1

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RECURSIVE: FACTORIAL

ACTIVITY: FACTORIAL

Similar to the sum but multiplication this time

factorial(0) = 0

factorial(1) = 1

factorial(4) = 1*2*3*4 = 24

factorial(5) = 1*2*3*4*5 = 120

Lets take a look at the iterative approach

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n = 3

```
def factorial(3):
    if 3 == 1: return 1
    else: return factorial(3 - 1) * 3
```

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n = 3

```
def factorial(3):
    if 3 == 1: return 1
    else: return factorial(3 - 1) * 3
```

2

```
def factorial(2):
    if 2 == 1: return 1
    else: return factorial(2 - 1) * 2
```

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```

2

```
def factorial(2):
    if 2 == 1: return 1
    else: return factorial(2 - 1) * 2
```

1

```
def factorial(1):
    if 1 == 1: return 1 ✓
    else: return factorial(1 - 1) * 1
```

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n = 3

```
def factorial(3):
    if 3 == 1: return 1
    else: return factorial(3 - 1) * 3
```

2

```
def factorial(2):
    if 2 == 1: return 1
    else: return 1 * 2
```

```
def factorial(1):
    if n == 1: return 1 ✓
    else: return factorial(1 - 1) * 1
```

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n = 3

```
def factorial(3):
    if 3 == 1: return 1
    else: return 2 * 3
```

= 6

```
def factorial(2):
    if 2 == 1: return 1
    else: return 1 * 2
```

```
def factorial(1):
    if n == 1: return 1 ✓
    else: return factorial(1 - 1) * 1
```

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RECURSIVE: FIBONACCI

EXAMPLE: FIBONACCI

a series of numbers where each number (Fibonacci number) is the sum of the two preceding ones, usually starting from 0 and 1

```
fibonacci(0) = 0
fibonacci(1) = 1
fibonacci(4) = 0+1+1+2     = 3
fibonacci(5) = 0+1+1+2+3   = 5
```

Lets take a look at the iterative approach

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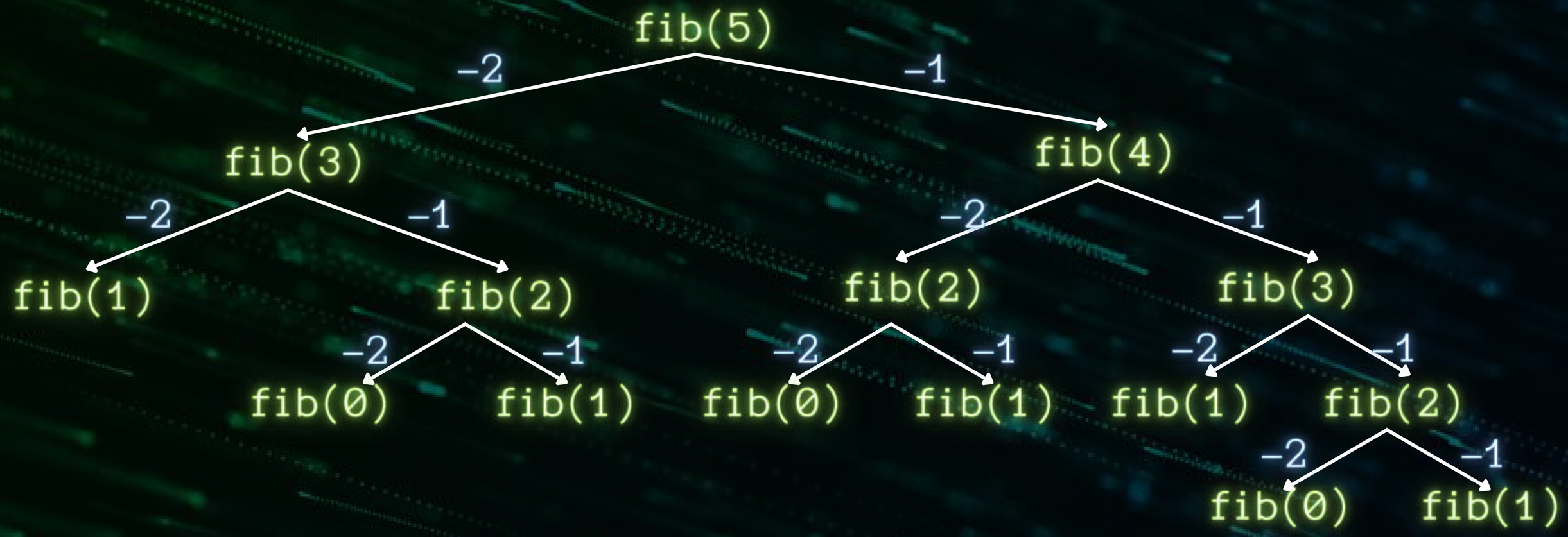

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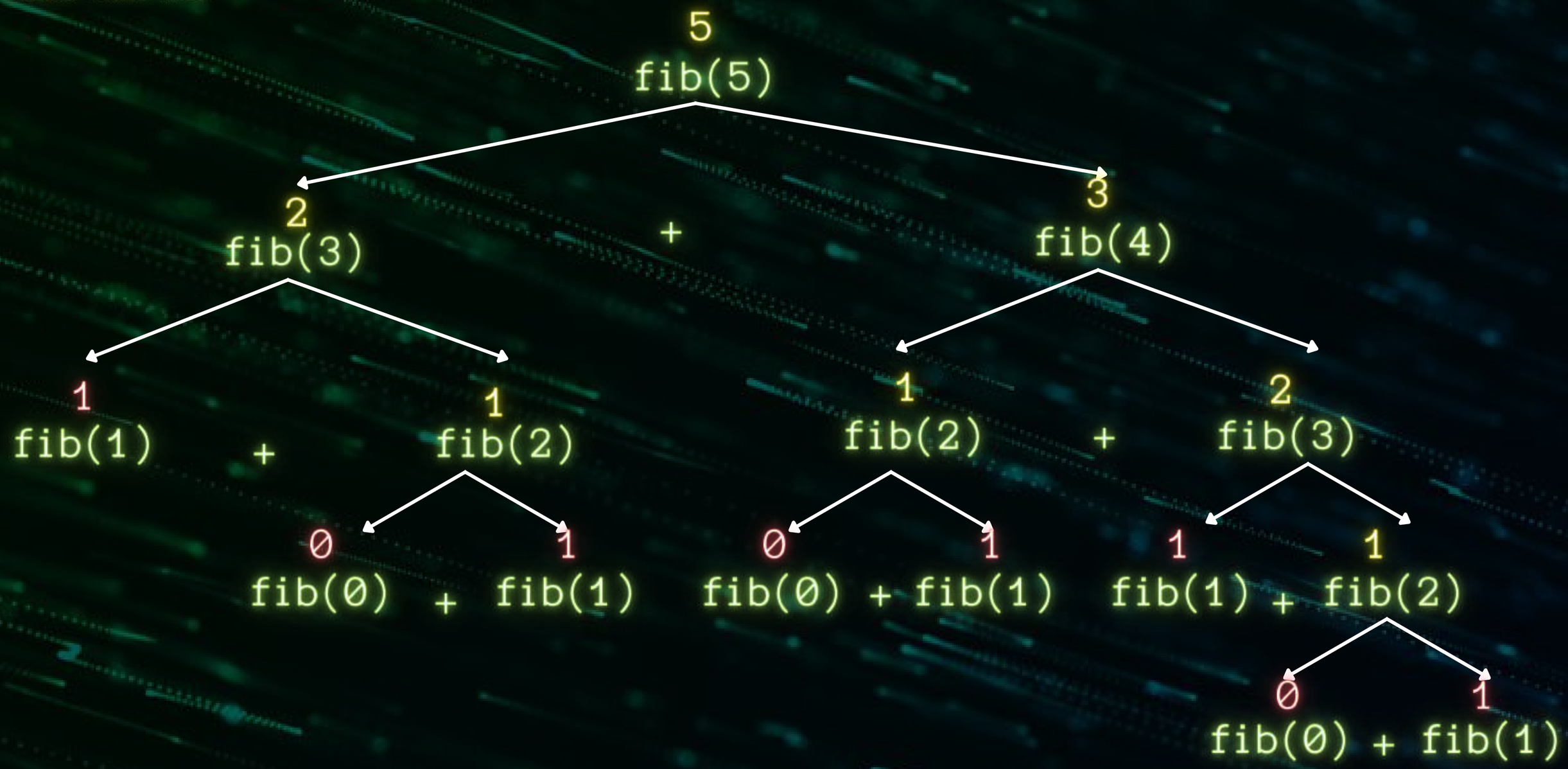

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WHY USE RECURSIVE?

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WHY USE RECURSIVE?

We use recursion for Trees and Nested Structures

1. The "Unknown Depth" Problem

Imagine you are writing a program to find a specific file on your computer.

- Level 1: C:/
- Level 2: C:/Users/
- Level 3: C:/Users/Von/
- Level 4: C:/Users/Von/Documents/
- ...

You don't know if the file is 3 levels deep or 50 levels

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WHY USE RECURSIVE?

```
for folder in drive:
    for subfolder in folder:
        for file in subfolder:  # What if the file is deeper?
          if file == "secret.txt":
          print("Found it!")
```

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WHY USE RECURSIVE?

```python
def find_file(folder, target):
    for item in folder:
        # 1. Base Case: It's a file. Check it.
        if is_file(item):
            if item.name == target:
                print("FOUND IT!")

        # 2. Recursive Case: It's a folder. Dive in!
        elif is_folder(item):
            # This function calls itself to handle the new depth
            find_file(item, target)
```

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ACTIVITY: STRING REVERSAL

How do you reverse "HELLO" without a loop? You take the last letter 'O' and add it to the reverse of "HELL".

- Recursive Step: Take the last character text and add it to the result of reversing everything else text.
- Base Case: The simplest input is nothing.

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ACTIVITY: THE PYRAMID BUILDER

Can you print a triangle of stars using recursion? The order of your code lines matters!

- Step 1: Call build_pyramid(n - 1) (Go deeper first!).
- Step 2: Print "*" multiplied by n.
- Base Case: If n is 0, return (stop).

Expected Output:

```
*
**
***
****
```

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ACTIVITY: NUMBER SUM

Can you calculate the sum of digits (e.g., 452 → 4+5+2 = 11) using only math? No converting to Strings allowed!

- Step 1: Get the Last Digit using modulo (n % 10).
- Step 2: Add that digit to the result of sum_digits called on the Rest of the Number (n // 10).
- Base Case: If n is 0 (no digits left), return 0.

Expected Output:
print(sum_digits(452)) should output 11